

Project Report

Preparing and transforming Superstore sales data with Python

1. Introduction

This project focuses on cleaning, transforming, and analyzing a sales dataset (`store.csv`) in order to generate meaningful business insights for decision-making.

The analysis is strictly based on the **sales** column, as the dataset does not contain profit, cost, or margin information. Therefore, the study concentrates entirely on revenue-based metrics.

The workflow follows a structured data engineering pipeline:

1. Data ingestion
2. Data cleaning
3. Feature engineering (transformation)
4. Distribution analysis & anomaly detection
5. KPI calculation
6. Export of the clean dataset

2. Data Ingestion

The dataset was loaded using the **pandas** library.

Main functions used:

- `read_csv()` → to import the dataset
- `Head()` → to preview the first rows
- `info ()` → to inspect data types and structure
- `shape` (attribute) → to check the number of rows and columns

This step allowed us to understand the dataset structure before modifying it.

3. Data Cleaning

The goal of this stage was to prepare the dataset for analysis.

3.1 Standardizing Column Names

We applied string transformation methods on column names:

- `str.strip()` → remove extra spaces
- `str.lower()` → convert to lowercase
- `str.replace()` → replace spaces and hyphens with underscores

This ensures compatibility with SQL, better readability, and avoidance of future errors.

3.2 Date Conversion

The columns `order_date` and `ship_date` were converted to datetime format using `to_datetime()` function.

This transformation is essential for time-based calculations.

3.3 Handling Missing Values

We inspected missing values using `isnull()` and `sum()` functions.

Missing postal codes were handled using `fillna()`

3.4 Removing Duplicates

Duplicate rows were removed using `drop_duplicates()`. This prevents double-counting in KPIs.

4. Feature Engineering (Transformation)

New derived features were created to enrich the dataset and improve analysis.

4.1 Shipping Delay

A new column was created to measure delivery performance by calculating the difference between ship date and order date.

Functions and attributes used: Date subtraction and `.dt.days`

This produced the **shipping_delay** metric (in days).

4.2 Time-Based Features

From the order date, we extracted: Year → using `.dt.year`, Month → using `.dt.month`, and Year-Month → using `.dt.to_period("M")`

These features allow time-series and growth analysis.

5. Distribution Analysis & Anomaly Detection

5.1 Descriptive Statistics

To understand sales distribution, we used: `describe()`

This provided: Mean, Minimum, Maximum, Standard deviation, and Quartiles.

5.2 Negative Value Detection

We checked for invalid values using: Conditional filtering (`df["sales"] < 0`), and `sum()`. This ensures data integrity.

5.3 Outlier Detection (IQR Method)

We applied the Interquartile Range (IQR) method using `quantile(0.25)`, and `quantile(0.75)`

Then calculated:

- $IQR = Q3 - Q1$

Boundaries were defined using the standard $1.5 \times IQR$ rule.

Outliers were identified using conditional filtering and flagged using `numpy.where()`

Instead of removing outliers, we created an indicator variable :
`is_sales_outlier`

This preserves data while allowing analysis of extreme values.

6. KPI Calculation for Commercial Director

We focus exclusively on **sales-based KPIs**.

6.1 Revenue Metrics

- Total Revenue → using `sum()`
- Average Sale → using `mean()`

6.2 Aggregated Performance Metrics

We used:

- `groupby()`
- `sum()`
- `sort_values()`
- `head()`

To calculate:

- Sales by category
- Sales by region
- Sales by segment
- Top 10 products by revenue

6.3 Monthly Growth Rate

Monthly sales were calculated using `groupby()` on the year-month variable.

Growth rate was computed using `pct_change()`

This measures the percentage variation between months.

7. Exporting the Clean Dataset

The final dataset, including new features and anomaly flags, was exported using `to_csv()`.

This dataset is ready for :

- SQL database integration
- Dashboard creation
- Further business analysis

8. Important Note on Profit and Margin

This dataset contains only the **sales** variable.

It does not include:

- Cost
- Profit
- Margin

Therefore, it is not possible to calculate:

- Profitability
- Profit margins
- Cost-based ratios

Such metrics can only be computed if the cost data is available, or a profit margin is assumed (which would introduce hypothetical values).

Since the objective was to perform a reliable and data-driven analysis, no assumptions were made beyond the available data.

The analysis remains strictly revenue-based.

9. Conclusion

This project successfully transformed a raw dataset into a structured, clean, and enriched analytical dataset.

Through:

- Data cleaning

- Feature engineering
- Distribution analysis
- Anomaly detection
- KPI calculation

We produced actionable insights based solely on sales performance.

The dataset is now ready for advanced analytics, SQL integration, or business intelligence visualization.